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Dr Ben Metcalfe
Head of the Department of Electronic & Electrical Engineering
Faculty of Engineering and Design
University of Bath
Claverton Down, Bath BA2 7AY, United Kingdom

Dear Dr Metcalfe and Members of the Selection Committee,

Application for Lecturer in Electronic & Electrical Engineering, Reference SS13726

I am writing to apply for the post of Lecturer in the Department of Electronic & Electrical Engineering. My research concerns the **integration of AI/ML into electronic and electrical engineering**, specifically deep-learning methods that accelerate the multi-physics simulation workflows behind electric machine and power-conversion design. I am currently a Research Assistant at the Power Electronics and Machines Centre (PEMC), University of Nottingham, having completed my PhD in Electrical Engineering at Southeast University (SEU), Nanjing, in September 2025.

My research record includes **22 journal papers (17 in IEEE Transactions)**, ten as first author, 18 peer-reviewed conference papers and 9 patents, recognised by the **2025 IEEE TEC Electric Vehicle Design Competition (1st Prize)** and the **2025 IEEE Transactions on Transportation Electrification Prize Paper Award (2nd Prize)**. The record maps directly onto the research directions named in the advert. On **using AI/ML to accelerate traditional computational workflows**, my ICEM 2026 paper develops an implicit-physics deep-learning surrogate that predicts the full magnetic field of an interior PM machine with $R^2 = 0.9996$ for torque and more than 94% inference speed-up over finite-element analysis, on low-cost GPU hardware; a companion physics-grounded optimisation framework cuts a 67-hour FE optimisation to 2.3 hours. On **data-centric approaches**, my work builds structured FE design datasets and includes a patented method for high-precision machine performance databases. These methods stand on an in-house Python finite-element code package I develop, and on dynamometer-validated industrial prototypes: through a multi-year collaboration with Inovance, a world-leading EV traction-machine supplier, I delivered two high-speed oil-cooled traction machines (18,000 rpm, 113 kW; 20,000 rpm, 270 kW) through the full industrial cycle.

My five-year plan, set out in the attached research statement, develops this into a programme on **AI-native simulation and design of electromagnetic energy-conversion devices**, extending the learned simulator toward general material topologies, eddy-current and PWM-driven conditions, and differentiable design. It aligns with **The Foundry: Centre for Digital, Manufacturing & Design** and serves sustainable transport electrification. I would pursue an **EPSRC New Investigator Award** within the first eighteen months, then Innovate UK and industrially co-funded projects building on my prototype delivery record. My current role on the multi-institution AIMD programme (Nottingham, Bristol, Newcastle) has shown me how large collaborative UK programmes run, and my co-author network already includes the University of Bath through joint work with colleagues associated with IAAPS.

On teaching, I served as teaching assistant on the SEU postgraduate course *Integrated Practice of Electrical Machine Systems*, and my research practice supplies ready material for **applied computer systems engineering and AI** teaching: scientific programming and numerical methods through the in-house FE package, embedded C/C++ on **TI C2000 and STM32 motor-drive platforms**, and application-focused contributions to the **postgraduate AI course** drawn from deploying deep learning on physical systems. I propose a final-year unit on *AI for Engineering Simulation and Design* with computational classes built on working course infrastructure, as detailed in my teaching statement. I would undertake the **PHEAF** programme on appointment, with HEA Fellowship as an early priority.

My CV, research statement and teaching statement are attached. I would be glad to discuss my application further at interview.

Yours sincerely,

Wentao Zhang